# Forest Rail

The Jingzhang Railway Heritage Park already left the rails in the city: people run, push strollers, walk along the tracks. This proposal does not reinvent that park, and does not turn it into a tech showcase. It asks something narrower — how the Dazhongsi end connects into the park, where people go after exiting the metro, where they can stop, whether they can sit down and work quietly, how seats and light strips can be safely AI-regulated, and how daily essentials can enter public space without a brand-store matrix [source:AGENT-TASKBOOK].

The answer is Forest Rail:

• Forest: Treat the Dazhongsi section as a park — manicured lawn, spaced trees, canopy above rooftops. Look out and see trees, not the glass wall opposite. The park is where people quiet down; human–nature relationship comes first.

• Rail (two tracks, separated): Jingzhang heritage rails are the urban clue. Decorative guiding rails give direction; AI operating rails run autonomous shuttles for passengers + automatic cargo configuration, safety first; physically separated from walking and decorative rails.

• Forest Rail terminal: intelligently regulates all public open seats / shared workstations, public ground light strips / low-illuminance edge lights, and daily-essentials supply at unbranded points (open/close, restock, rebalance). No face recognition, no brand ads.

• Unbranded daily-essentials trial points: do not start with each brand opening its own storefront in every direction. Instead, like old food shops — unified fronts, no brand logos; trial many sites; AI algorithms find where fit is highest and where one person's daily essentials can be met; high-fit nodes later open formal stalls; elsewhere use community quick-turnover shops, with AI inventory and autonomous cargo delivery.

> One line: rails are the city's clue; the park lets people quiet down; rusted rails point the way; autonomous shuttles put safety first; seats, light strips, and daily essentials can be smart-regulated — look out and see nature.

This proposal carries forward the Jingzhang Railway spirit of daring breakthrough — turning heritage rails into walkable, sittable, shared public infrastructure — without empty empowerment slogans or viral landmarks.

What this proposal refuses: district-wide building control, viral landmarks, light shows, planting strips posing as parks, private pods disguised as shared desks, decorative rails treated as climbing gear, “smart” traded for surveillance, brand-store matrices flooding the park.

Only one rail-guided corridor is added (guiding rails primary; operating rails only on isolated conceptual segments). Outside it, the existing street network stays.

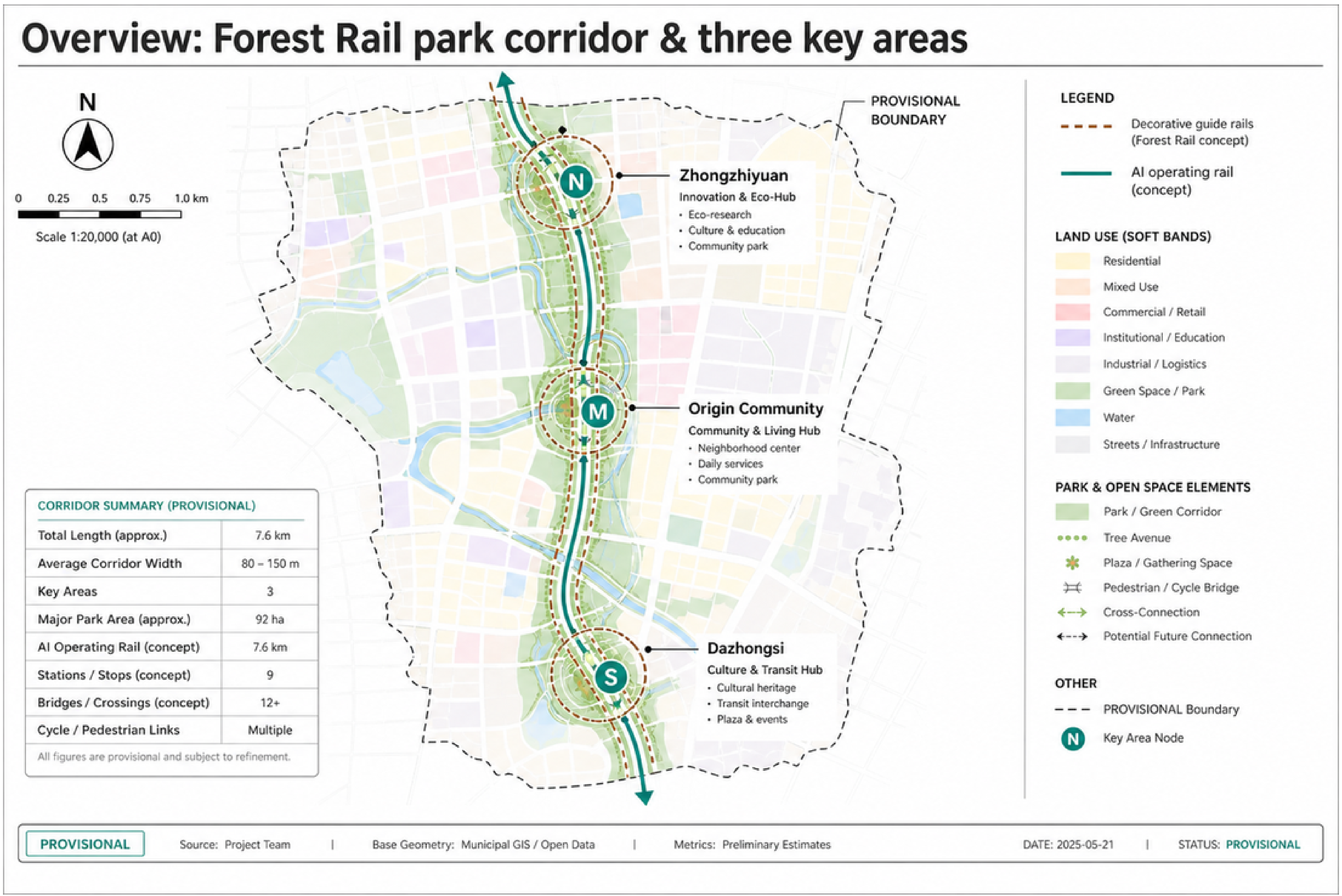
## Design Basis and Source List

This proposal follows the Pre-qualification Announcement for the Centennial Jingzhang AI Innovation Belt Urban Design International Open Call [source:OFFICIAL-ANNOUNCEMENT], with machine-readable references from `brief/site-package/` [source:SITE-PACKAGE].

Current spatial data uses provisional boundaries (`geometry/site_boundary.geojson`, `geometry/key_areas.geojson`), labeled `provisional_constraint`, `official_boundary=false`. All metrics and drawings must be recalculated when official boundaries are released. This limitation does not block content scoring, but spatial conclusions in this text are for design discussion only.

The source registry (`data/source_registry.json`) lists 7 formal sources, 1 background source, and 1 provisional-only source. Background and provisional sources are not elevated to statutory authority [source:SOURCE-REGISTRY].

Authority order in this package: GeoJSON → metrics → matrices → sources/assumptions/self_check → proposal.md → figures → HTML → PDF. Concept renders do not replace evidence diagrams.



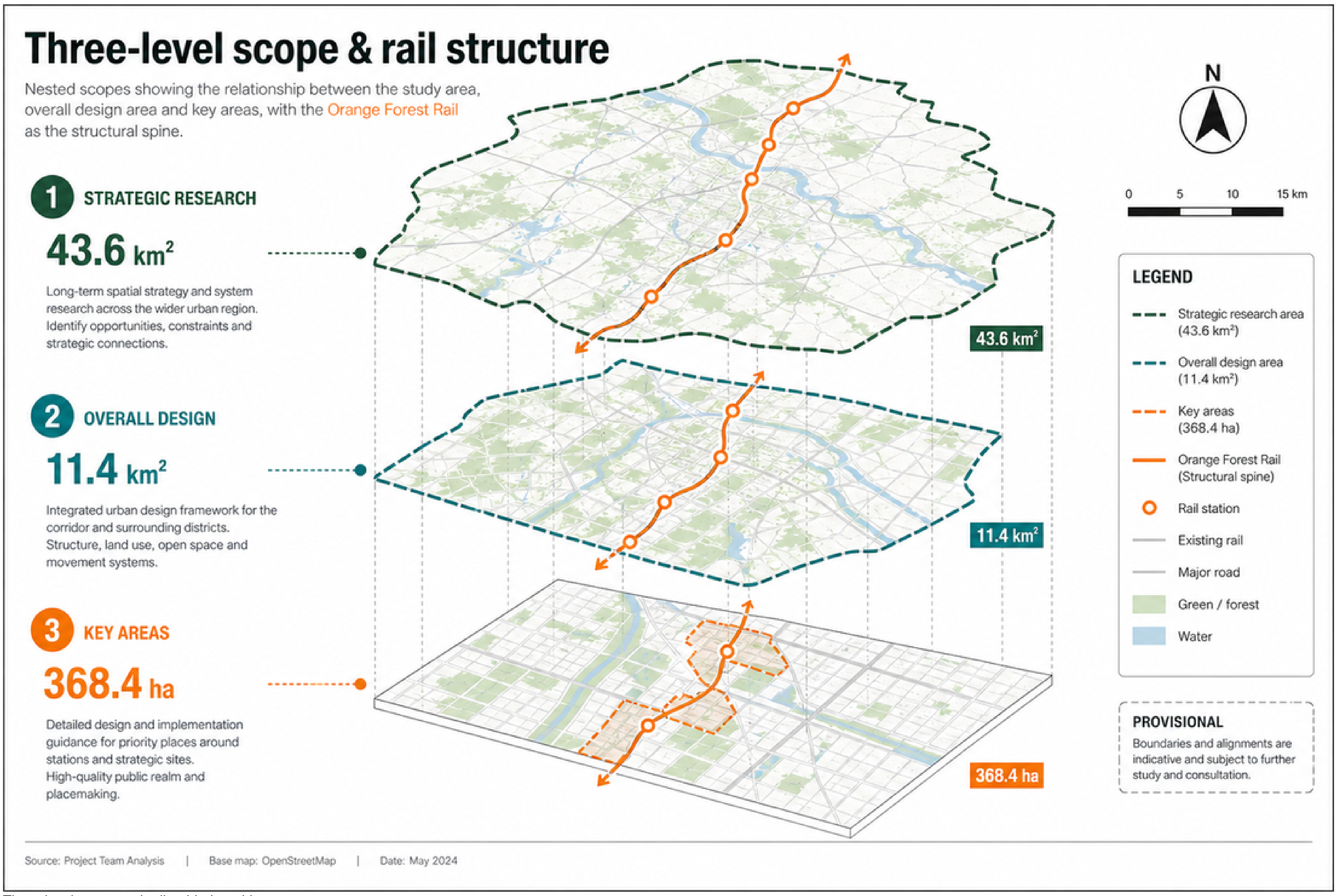

Source evidence chain and submission package diagram

## Three-Level Scope Framework

The proposal follows the three tiers specified in the announcement [source:SITE-PACKAGE]:

| Tier | Area | What Forest Rail does here |
|------|------|---------------------------|
| Strategic research (43.6 km²) | AI industry ecosystem and urban morphology | Confirms the Jingzhang corridor as spine; Forest Rail can extend north from Dazhongsi |
| Overall design (11.4 km²) | Urban renewal around the heritage park | Rail corridor links three key areas; park walking and soft paving; canopy as character control |
| Key area design (368.4 ha) | Three detailed design zones | Dazhongsi completes the park set: canopy, rail path, soft paving, shared desks, rust |

Tiers are linked: strategy sets direction, overall design lands layers, key areas test whether components fit parcels.



Three-level scope and rail-guided corridor

## Coordinated Research Area: Industry and Future City Research

The Jingzhang Railway is the narrative already embedded here. It opened in 1909; after going underground in 2019, the surface alignment became a heritage park corridor. The announcement requires integrating centennial Jingzhang culture, Zhongguancun innovation culture, and AI innovation culture [source:AGENT-TASKBOOK].

Forest Rail's judgment is narrow: do not invent a new story — use the rails; do not invent a new green corridor — connect the park to everyday Dazhongsi. Rails stay on the ground and rust; people walk soft paving between them. The line connects three key areas — Zhongzhiyuan by Qinghe, Origin Community near universities, Dazhongsi by the metro. The rail-guided corridor is the spine, not decoration.

Industry links follow routes people actually walk:

• University sourcing: work moves south along the heritage corridor, not floating circles

• Open-source collaboration: Origin Community hosts release and code work

• Enterprise translation: Dazhongsi hosts agents, terminals, and data-element firms

• Public services: ask-a-person nodes along the rail path, not unmanned screen walls

• International visits: walkable rail-path itineraries, not parade stages

Naming: Forest Rail is both title and spatial description. No logo needed — the track is the mark. Rust and canopy green are enough.

### Urban Parks and Linear Parks: Global / Domestic Cases (Transfer Boundaries)

At the strategic tier, Forest Rail does not invent a new green-corridor story. It aligns transferable mechanisms from well-known urban and linear parks to the Jingzhang heritage corridor. Cases below are publicly built or operating projects. This proposal learns publicness and linear organization — not viral installations or climbable heritage consumption.

| Case | Transferable mechanism | Boundary for this proposal (learn / do not learn) |
|------|------------------------|---------------------------------------------------|
| New York Central Park | Multi-entry + layered walking; park as a public right | Learn: treat Dazhongsi as a real park (open lawn / spaced trees / seating); Do not: mega single-greenfield or closed heritage management |
| New York High Line | Abandoned rail → linear walk; keep rail artifacts as memory | Learn: rail-reuse narrative; Do not: elevated tourist consumption or climbable rails; our decorative rails ≠ climbing — soft paving for feet |
| Philadelphia Rail Park | Phased opening; 10–15 minute community reach | Learn: one-block-at-a-time lightweight pilots; Do not: sell unfinished segments as finished, or copy heavy elevated steel |
| London King's Cross public realm | Station squares stitching walking and mixed uses | Learn: exit into public space (Dazhongsi four quadrants); Do not: mall-led plaza logic |
| Madrid Río | Infrastructure gives way to continuous riverside green | Learn: stitch blocks split by infrastructure; Do not: tunnel-scale cost and schedule |
| Seoul Cheonggyecheon | Uncover water/green; everyday linear civic use | Learn: linear publicness; Do not: pumped faux-ecology or massive excavation narratives |
| Shanghai Xuhui Runway Park | Transport relic → park beside street | Learn: clear remnant-to-park narrative; Do not: wide hard runway feel or photo-booth stacking |
| Beijing Olympic Forest Park | Urban green heart; all-ages lawn and canopy | Learn: urban-forest character and inclusion; Do not: national-axis scale or a new mega-park land take |
| Jingzhang Railway Heritage Park (existing) | Rails already in the city; "three paths + green" walking | Learn: connect into Dazhongsi as incremental link; Do not: redo the park or invent a parallel corridor |

Stance: carry forward Jingzhang daring — move heritage rails from "to be seen" toward "to guide, to share work, to host conceptual short-haul" — with the breakthrough landing on safety separation and everyday publicness, not empowerment slogans or viral gear.

Park case background sources are registered in `sources.json` (not statutory). North American urban/linear parks: [source:CASE-CENTRAL-PARK], [source:CASE-HIGH-LINE], [source:CASE-RAIL-PARK]. Europe and East Asia: [source:CASE-MADRID-RIO], [source:CASE-CHEONGGYECHEON]. Domestic and local baseline: [source:CASE-XUHUI-RUNWAY], [source:CASE-OLYMPIC-FOREST], [source:CASE-JZ-HERITAGE-PARK].

## Overall Design: Rail-Guided Corridor and Urban Renewal

### Corridor Structure

The rail-guided corridor follows the Jingzhang heritage park alignment, with Dazhongsi as the southern gateway. It is not a newly drawn boundary — it reuses the railway alignment. The park is already walked; this proposal mainly connects Dazhongsi Station into that park.

```
Zhongzhiyuan (N) ──rail── AI Origin Community (M) ──rail── Dazhongsi (S) │ │ Qinghe interface Campus-district walking link
```

Tracks along the corridor serve two purposes and are physically separated [source:AGENT-TASKBOOK]:

| Purpose | Material | Function | Boundary |
|---|---|---|---|
| Decorative guiding rails | Rusted / replica rails / ground painting + soft paving | Direction: eyes follow rust; feet on timber or crushed stone; sittable, slow | No running or climbing on rails; no light strips on decorative rails; signage minimal (Mode A) |
| AI operating rails (concept) | Autonomous shuttles on isolated operable segments | Low-speed passenger carriage + automatic cargo/materials configuration; safety first (E-stop, speed limit, yield to walking, fail-safe stop) | Separated from walking and decorative rails; not commercially mature; subject to professional study |

### Forest Rail Terminal: Environment Regulation (Seats · Light Strips · Daily Essentials · Access)

The Forest Rail terminal is an environment and daily-essentials regulation system — not building-wide control, not ad screens:

| Target | What AI does | Safety & privacy boundary |
|---|---|---|
| Public open seats / shared desks | Open/close and power by time, noise, temperature/humidity, rain, anonymous crowd density | No identity logging; prefer close/lower power when crowded or weather is bad |
| Public ground light strips / low-illuminance lights | Brightness and zone on/off by daylight, time of day, foot traffic; edges and seating only | No light shows, projections, or colored chasing lights; fault defaults to off or ultra-dim |
| Node access | Quiet workstations, rain-shelter seats, appointment safety nodes by schedule | Offline fallback to human/paper rules; human override always available |
| Autonomous shuttles (concept) | Dispatch, auto cargo config (including restock for daily points), path yielding | Safety first: any collision risk, boundary breach, or E-stop failure → stop and restore walking priority |
| Unbranded daily-essentials trial points | Open/close, restock, and rebalance daily necessities by anonymous footfall, time of day, stockouts, and turnover; after multi-site trials, output fit scores | No brand logos, no face recognition, no ads; formal stalls only after human review; failed sites can be withdrawn |

Most stretches are decorative guiding rails only. A few isolated stretches conceptually retain AI operating capacity. Pedestrians have priority at crossings; operating rails yield. If operating rails fail or are never deepened, guiding rails, seat regulation, light-strip regulation, and daily-essentials regulation still stand on their own.

## Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Renewal within the overall design scope falls into three categories:

- Retain: Existing quality buildings along the heritage corridor — unchanged
- Renovate: Buildings needing ground-floor program and facade adjustment — remove LED screens, increase under-canopy transparency
- Renew: New construction on identified low-efficiency sites — building height controlled below tree canopy

Land use classification follows national standards [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE], covering the design boundary without overlap [data:geometry/land_use.geojson#LU-001]. Building footprints and retain/renovate/renew classification are recorded in [data:geometry/buildings.geojson#BLDG-001].

Where FAR, building height, or road setbacks lack official data, they are marked as unknown. No numbers are fabricated.

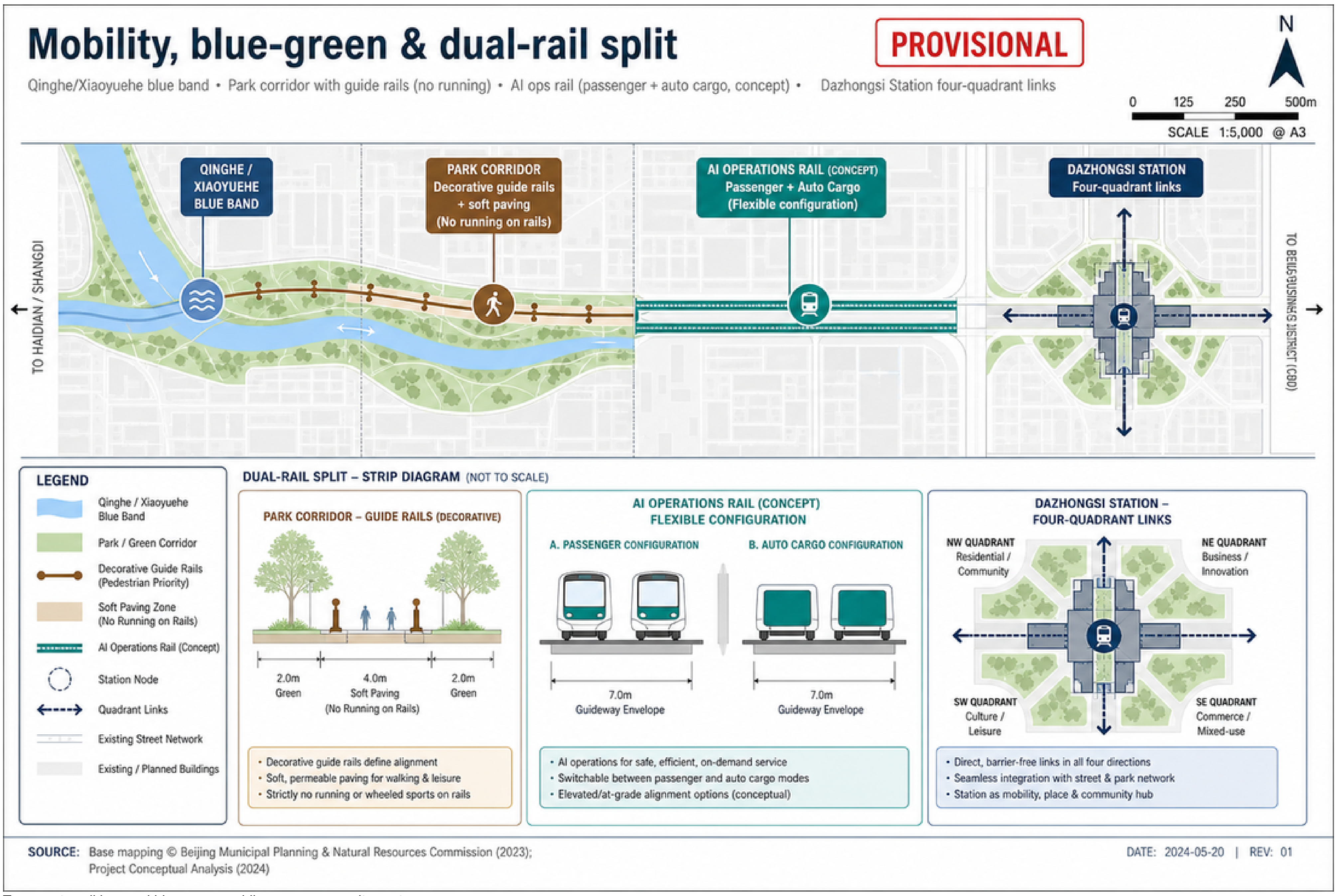
## Transport, Rail, Municipal Infrastructure, and Public Services

Transport responds to the announcement's requirements for rail station integration, road micro-circulation, walking network gaps, and green transport [depth:traffic_rail_slow_parking].

Key interventions:

1. Dazhongsi Station four-quadrant pedestrian connection: Metro Line 13 Dazhongsi Station — establish walking connections across all four quadrants of the intersection, with rail paths extending from station exits into the district 2. North Fifth Ring crossing node: Pedestrian linkage where the heritage corridor crosses the North Fifth Ring Road (underpass or overhead facility) to maintain north-south continuity 3. Rail path / municipal road intersections: Pedestrians have priority at crossings; rail tracks embedded in road surface indicate direction; no traffic signals govern the rail path

Road and walking network layers remain within the submission boundary [data:geometry/roads.geojson#ROAD-001], cross-checked with public space and green space. With a provisional boundary, transport conclusions serve as design discussion only.




Transport, walking, and blue-green public space composite system

Municipal infrastructure covers edge-computing nodes, distributed energy, and traditional utility integration. Missing utility, energy, and drainage data are listed as prerequisites for detailed design, not fabricated.

## Blue-Green Network, Public Space, and Urban Character

Blue-green space follows the Jingzhang heritage corridor as its spine, coordinating Qinghe River, Xiaoyue River, and surrounding community movement [depth:blue_green_public_space].

The tree canopy layer is the primary character control mechanism: native trees with crown spread ≥8m (Chinese Scholar Tree, ash, ginkgo, goldenrain tree), targeting ≥70% coverage of building rooftops. Trees are not pruned into geometric shapes. Building height below the canopy — this is the basic criterion for the urban forest.

Character principles:

- Building facades use materials that age: brick, concrete, wood. No smooth glass curtain walls
- Rust and grey-green are the base colors, derived from oxidized rail and tree canopy
- Rooftops are for greening or equipment, not sculptural form
- Public art accepts only site-history-related, aging-compatible works — iron, stone, wood. No stainless steel or lighting installations

Which elements are design recommendations versus those pending official confirmation is itemized in `standard_matrix.json`.

## Key Area Detailed Design

### Zhongzhiyuan AI Autonomous Innovation Accelerator

Location: Northern end, along Qinghe River [data:geometry/key_areas.geojson#PROV-KEY-001].

Position: Garden-type full-stack autonomous innovation district. The Qinghe interface is its public living room — combining green space, stormwater management, walking and cycling, and AI safety testing display.

Spatial actions:

- Qinghe interface: Set back buildings, create under-canopy walkways and waterfront grass slopes
- Industry display: Standards development, safety evaluation, model red-team testing translated into visitable collaboration nodes — by appointment, not as open attractions
- External transport: Strengthen walking connections to areas north of the Fifth Ring
- Low-carbon energy: Distributed PV combined with edge computing, as an infrastructure prototype for further development

### Beijing AI Origin Community

Location: Middle section, near Tsinghua, Beihang and other universities [data:geometry/key_areas.geojson#PROV-KEY-002].

Position: University-adjacent technology transfer and talent community.

Spatial actions:

- Campus-park-district walking integration: Break through walls and gaps, connect campus gates to park entrances via rail paths
- Launch space: Open-source release hall, public code wall (physical screen displaying real-time open-source contributions), small presentations
- Talent services: Housing, dining, sports embedded in the streetscape, not concentrated in a "talent apartment complex"
- Open-source collaboration: 24-hour open collaboration spaces, divided into quiet zones and discussion zones by usage intensity

### Dazhongsi AI Industry Cluster — Urban Forest

Location: Southern end, adjacent to Metro Line 13 Dazhongsi Station [data:geometry/key_areas.geojson#PROV-KEY-003].

Position: Urban intelligent economy district, and the fullest end of Forest Rail — exit the station into a park, and find a shared desk in that park.

Five components:

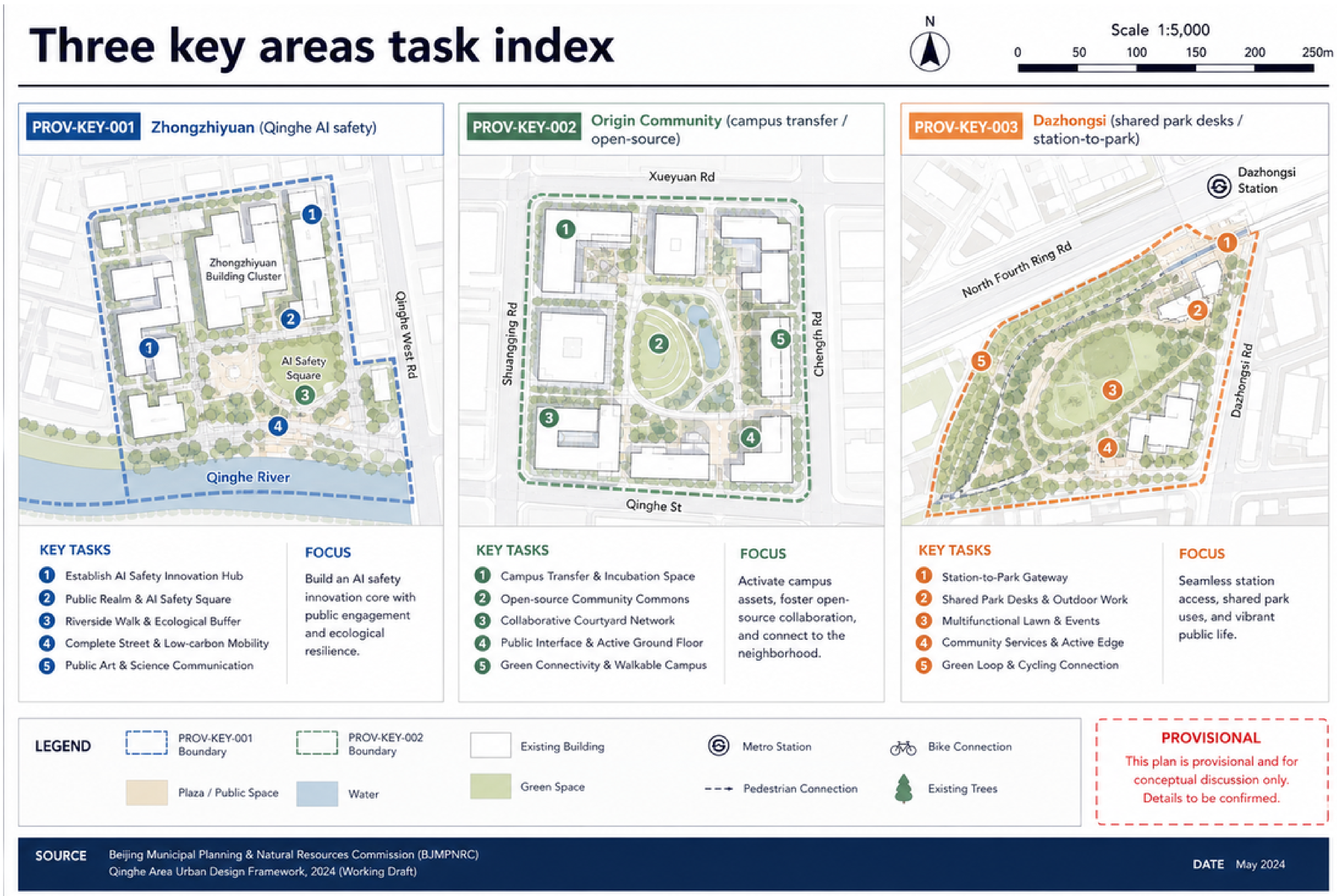
| Component | How it is used in Dazhongsi | What is not allowed |
|-----------|------------------------------|---------------------|
| Canopy layer | Chinese Scholar Tree and ash as primary species, crown spread ≥8m, covering existing building roofs and new low-rise structures | No geometric pruning; no ornamental small trees as substitutes |
| Rail path | From Dazhongsi Station exits, decorative guiding rails (rusted / replica / ground painting) point into the park; people walk soft paving | No elevated landscape bridges; no polishing; no painting; no light strips; no running or climbing on rails |
| Soft paving | Between-rail walkways use timber boardwalk, permeable brick, or crushed stone; comfortable underfoot | No large-area asphalt; no polished stone (slippery in rain) |
| Under-canopy workspace | Dedicated shared workstation precinct in the park: many desks, glass partitions, hot-desks anyone can use; building ground floors also shaded by canopy | No private assigned pods; no LED facades; no viral photo-booth cabins |
| Rust | Rail tracks and steel elements maintain natural oxidation as time markers | No Cor-Ten imitation rust panels; no artificial aging with clear coat |

Dazhongsi Station integration: From exits A/B/C/D, a rail path enters each quadrant. Emerging underground, the first view should be park and rusted rail — not ad screens.

Commercial services (unbranded daily essentials → formal stalls by fit): Do not start with each brand opening its own storefront in every direction. Instead, set unified, unbranded daily-essentials trial points (water, simple meals, basic daily goods — what one person needs in a day), like old food shops — unified fronts, no logos, no marquees. After multi-site trials, AI algorithms (anonymous footfall, time-of-day demand, turnover, stockouts) judge where fit is highest and where daily essentials can be met; high-fit nodes later open formal commercial stalls (open admission, not AI anointing a brand); elsewhere use community quick-turnover shops, with Forest Rail terminal AI managing inventory and autonomous shuttles delivering restocks. Company showrooms may face the rail path but must not overpower the park's quiet character; data-element services require compliance, authorization, and auditability.

Park integration: The rail path sits in a park — manicured lawn, spaced trees, soft paving. Shared desks and daily points both serve the park: rails are the urban clue; the park is where people quiet down and human–nature relationship comes first — not an office lobby or brand street moved outdoors.

Concept views (not formal evidence figures; spatial claims remain in GeoJSON / metrics / the five evidence diagrams):



Three key areas index and design tasks

All three key areas appear in `geometry/key_areas.geojson`. Current data is provisional constraint; the proposal, HTML, and self_check state that it cannot serve as approval or scoring basis.

# A Day Here: Personas and Scenarios

## Five User Personas

| Persona | Who | What they do on Forest Rail | Data boundary |
|---------|-----|----------------------------|---------------|
| Walking commuter programmer | AI engineer living nearby, walks to work daily | Exits Dazhongsi Station, walks 800m along the rail path to the office; eats lunch under the canopy; walks back along the rail path after work | No personal commute tracking; rail path foot traffic is anonymous counting only |
| Parent walking with child | Resident of surrounding neighborhoods | Pushes a stroller along the soft paving in the evening; child runs on soft paving beside the rails (not on the rusted rails); sits on under-canopy benches chatting | No family profiling; no commercial push notifications |
| Corporate visitor | Business guest of a major company | Arrives at Dazhongsi Station, rail path guides to company showroom; walks along rail path to next node after the meeting | Company logos and cases require rights clearance |
| Solo freelancer | Person without a fixed office | Finds an open shared desk in the park workstation precinct; opens when quiet, closes when crowded; hot-desk, no reserved seats | No recording of workstation user identity |
| Retired daily walker | 70-80 years old, lives nearby | Walks along the rail path every morning, sits on the bench at the curve watching trees; pavement must be non-slip | No health data collection |

## Scenario Cards (≥10)

| # | Scenario | Spatial carrier | What happens | Who uses it |
|---|----------|----------------|-------------|-------------|
| 01 | Exit station, enter park | Dazhongsi Station exits | Decorative guiding rails point to park entries; canopy provides auxiliary shade; rain relies on shelters/arcades as backup — not light shows | All arrivals |
| 02 | Rail path walking | Entire Dazhongsi rail path | Eyes follow rusted rails; feet on timber or crushed stone. Speed naturally slows. No destination signs urging you forward | Commuters, walkers |
| 03 | Shared under-canopy workstations | Dedicated park precinct with glass-partitioned multi-desk bays | Shared hot-desks: sit if free. Forest Rail terminal opens/closes seats and power by noise / climate / rain / crowd | Freelancers, lunch-breakers, passersby working briefly |
| 04 | Rusted rail curve bench | Rail path turning points | Benches on the outside of curves; sit and watch the rusted rail and trees. No signage; discover by sitting; terminal may schedule that seat group | Elderly, parents with children |
| 05 | Company showroom along the rail | Dazhongsi ground-floor retail | AI agent and terminal company display spaces open toward the rail path; visible as you walk past | Visitors, passersby |
| 06 | Rainy day under-canopy shelter | Dense canopy segments + node shelters | Canopy as auxiliary shade; heavy rain backed up by artificial shelters / semi-open arcades; AI prompts open seating and shelter nodes | Everyone |
| 07 | Open-source launch hall | Origin Community | Code contribution display and small presentations for universities and open-source communities. Physical screen scrolling real-time commits | Developers, university faculty and students |
| 08 | Inter-node autonomous shuttle (concept) | Isolated operable segments | Safety-first autonomous vehicles for passengers + automatic cargo; E-stop / speed limit / yield; separated from walking and decorative rails | Concept verification stage, not mixed daily use |
| 09 | Night light-strip regulation | Soft paving edges along the rail path | Terminal regulates public ground light strips / low-illuminance lights: edges and seating only. No light shows, no projections | Night runners, late returners |
| 10 | Qinghe innovation corridor | Zhongzhiyuan along Qinghe | Waterfront walkway connects to the rail path, stormwater grass slopes double as resting lawns. AI safety testing nodes open by appointment | Visitors, walkers |
| 11 | Data element salon | Dazhongsi district | Under compliance, authorization, and auditability, displays data element and digital asset circulation as an urban service interface | Enterprise clients, regulators |
| 12 | University tech transfer street | Origin Community | Incubation, display, legal, IP, and investment services organized along the rail path for university technology transfer | Startup teams, investors |
| 13 | Unbranded daily-essentials trial points | Multi-site fronts along the rail path | Unified food-shop-style fronts, no brand logos; AI regulates open/close, restock, and rebalance of daily necessities; multi-site trials output fit scores | Commuters, residents, passersby working briefly |
| 14 | High-fit nodes → formal stalls | Nodes with high fit after AI trials | Open formal commercial stalls (admission-based); elsewhere switch to community quick-turnover shops with AI inventory + autonomous cargo delivery | Community vendors, residents |

All AI governance follows data minimization, open sourcing, explainability, and human review. Urban intelligent agents assist in identifying walking network gaps, public space activity, and facility maintenance needs but do not replace planning approvals or output unauthorized personal profiles.

### Three Quiet Landmarks (Not Attractions)

No "check-in points" are designated here. The following are places you notice while walking — no signage, no wayfinding, no photo-taking recommendations:

1. Rusted rail curve with an old Scholar Tree: At a bend in the rail path, an existing large Scholar Tree. The curve makes you turn your head and see the tree. Unnamed, unmarked. 2. Rail path–Xiaoyue River junction, crushed stone square: Where the rail path crosses Xiaoyue River, a crushed stone surface forms a small square. Water, rusted rail, and canopy meet here. No sculpture. 3. Old platform retaining wall: If Jingzhang Railway-era platform remnants exist in the district (retaining walls, steps), they are kept as found. The wall surface is not cleaned; moss grows. A bench is placed beside it.

## Renewal Project List and Phasing

| ID | Project | Type | Key dependencies | Phase |
|----|---------|------|-----------------|-------|
| FR-01 | Dazhongsi Station four-quadrant rail path connection | Walking / rail integration | Rail station, intersection traffic organization | Near-term pilot |
| FR-02 | Rail path wayfinding segment installation (Dazhongsi) | Public space | Rusted rail sourcing, soft paving materials, drainage | Near-term pilot |
| FR-03 | Canopy layer planting (Dazhongsi) | Greening | Tree specification, soil improvement, utility avoidance | Near-term start, 3-5 year canopy maturity |
| FR-04 | Under-canopy workspace ground floor renovation | Building renovation | Ownership, ground-floor program adjustment | Medium-term |
| FR-05 | Zhongzhiyuan Qinghe innovation interface | Blue-green / industry | River blue line, ecology and flood control conditions | Medium-term |
| FR-06 | Origin Community campus walking integration | Walking | Campus boundary, wall removal negotiation | Medium-term |
| FR-07 | Inter-node AI operating rail concept segment (passenger short-haul + automatic cargo) | New infrastructure (concept) | Engineering feasibility, safety assessment, physical separation | Long-term / pending |
| FR-08 | Full rail path connection (three key areas linked) | Public space / walking | North Fifth Ring crossing, corridor ownership | Long-term |

Near-term pilots can start with lightweight interventions (spread crushed stone, place benches, plant saplings) without waiting for all engineering conditions. Medium and long-term projects require confirmed regulatory plans, municipal engineering, and property rights.

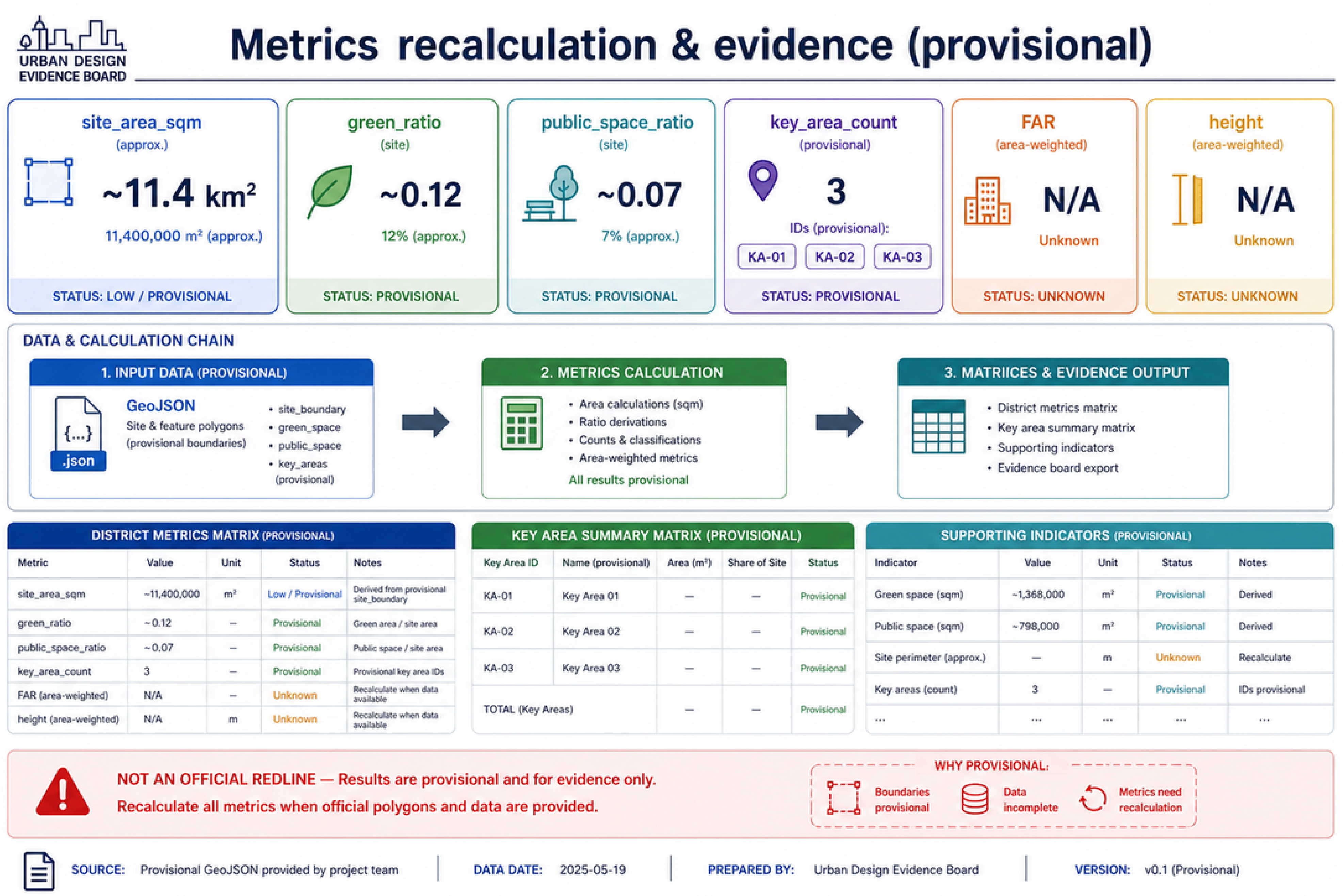
Phasing spatial evidence [data:geometry/phasing.geojson#PHASE-001].

## Metrics Framework

Metrics are divided into three categories:

| Type | Examples | Data source | Precision |
|------|----------|-------------|-----------|
| Spatial metrics calculable from submitted geometry | Boundary area, green ratio, public space ratio, building footprint | GeoJSON layers | Limited by provisional boundary |
| Control metrics requiring official regulatory plans | FAR, building height, building density, setbacks | Pending official plans | Currently marked unknown |
| Performance metrics requiring ongoing operational data | Rail path daily foot traffic, under-canopy workstation utilization, node access frequency | Post-operation collection | Currently design reference values |

Three categories enter `metrics.json`, `assumptions.json`, and `compliance_matrix.json` respectively. Full calculations are in `metrics.json`; this text explains design intent only.




Core metrics recalculation and evidence chain

Key spatial metrics can be verified against [metric:site_area_sqm] and [data:geometry/green_space.geojson#GREEN-001].

### Compliance Matrix Coverage

`compliance_matrix.json` maps each requirement from Announcement 1.3-1.5 and agent.1-agent.6:

| Requirement | Proposal section | Core evidence |
|-------------|-----------------|---------------|
| 1.3 World-class AI innovation ecosystem | Strategic research + Key areas | Innovation chain spatial layout, scenario cards |
| 1.4 Three-tier scope | Three-tier scope | Task table per tier |
| 1.5.3.1 Zhongzhiyuan | Key areas: Zhongzhiyuan | key_areas PROV-KEY-001 |

| 1.5.3.2 Origin Community | Key areas: Origin Community | key_areas PROV-KEY-002 |

| 1.5.3.3 Dazhongsi | Key areas: Dazhongsi | key_areas PROV-KEY-003 |

| agent.1 Overall concept | Overall design: Corridor structure | Rail-guided corridor + three key area linkage |

| agent.2 AI full-stack innovation | Strategic research + Zhongzhiyuan | Innovation chain, safety governance nodes |

| agent.3 AI+ scenarios | Personas and scenarios | 14 scenario cards |

| agent.4 Public space and landmarks | Dazhongsi urban forest + quiet landmarks | Five components + three quiet landmarks (not attractions) |

| agent.5 Cultural narrative integration | Strategic research: Why rails | Jingzhang Railway to rusted rail to Forest Rail |

| agent.6 Global activities and long-term operations | Renewal projects: Phasing | Near-term lightweight to medium-term renewal to long-term connection |

# Risk, Copyright, and Compliance

## Provisional Boundary

All spatial conclusions are limited by provisional boundary status. When official boundaries are released: re-run scaffold, self-check, drawings, and HTML. Single-file replacement is not sufficient.

## AI Operating Rail & Forest Rail Terminal (Concept)

• Autonomous shuttles: safety first — speed limit, E-stop, yield to walking and decorative-rail zones; stop on collision risk, boundary breach, or E-stop failure. May carry passengers and auto-configure cargo/materials pods (including restock for daily points). Conceptual suggestion, not a mature commercial claim.

• Forest Rail terminal: intelligently regulates public seats/shared desks, ground light strips / low-illuminance lights, and daily-essentials open/close/restock/rebalance at unbranded points; no face recognition, no brand ads; human override when offline.

• Unbranded daily-essentials trial points: start with no brand logos; AI outputs fit scores and supply suggestions only; upgrading to formal stalls requires human review and open admission; failed sites can be withdrawn. No brand-store matrix flooding the park.

• Decorative guiding rails: no climbing or running; physically separated from operating rails. Rails are the urban clue; the park prioritizes quiet and human–nature relationship.

## Missing Data

Gaps listed in `missing_data_checklist.csv` — official boundaries, regulatory plans, roads, municipal engineering, heritage protection — enter `assumptions.json` and self-check. Conclusions lacking official conditions are downgraded to pending confirmation.

## Copyright

The proposal provides bilingual versions (this file and `proposal.md`). Images, data, and code assets are documented in `sources.json` and `report/copyright_statement.md`. HTML pages do not load remote scripts, map tiles, fonts, iframes, or external APIs.

## Disclaimer

This proposal does not claim official approval, regulatory plan status, final property rights, final construction scale, or implementation guarantee.

# References

See [source:OFFICIAL-ANNOUNCEMENT], [source:AGENT-TASKBOOK], [source:SITE-PACKAGE], `sources.json`, `metrics.json`, `compliance_matrix.json`.


• brief/public-brief.md

• brief/site-package/design_brief.json

• brief/site-package/allowed_design_space.json

• data/processed/agent_fact_pack.md

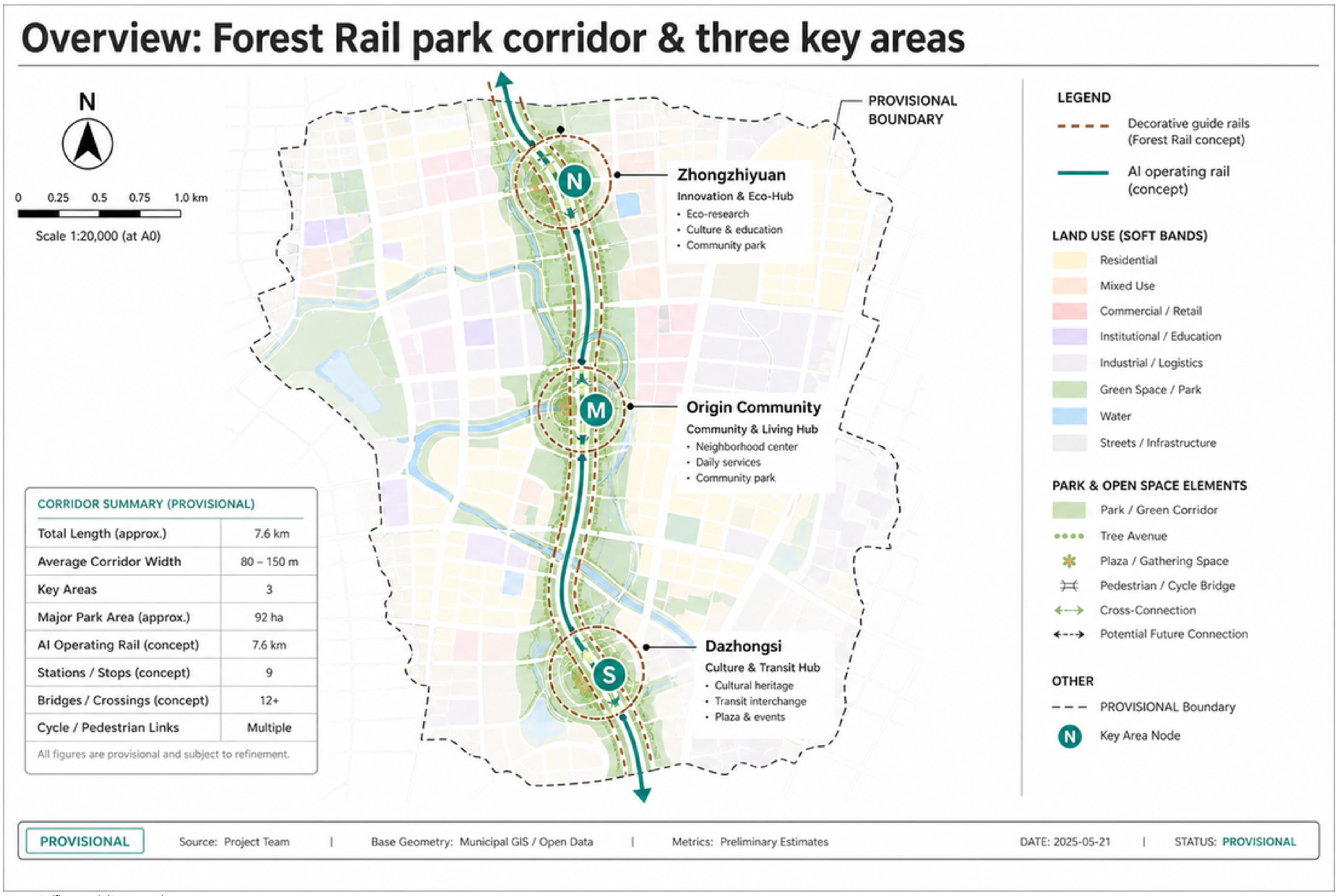
• data/processed/project_scope_summary.csv

• data/processed/agent_task_requirements.csv

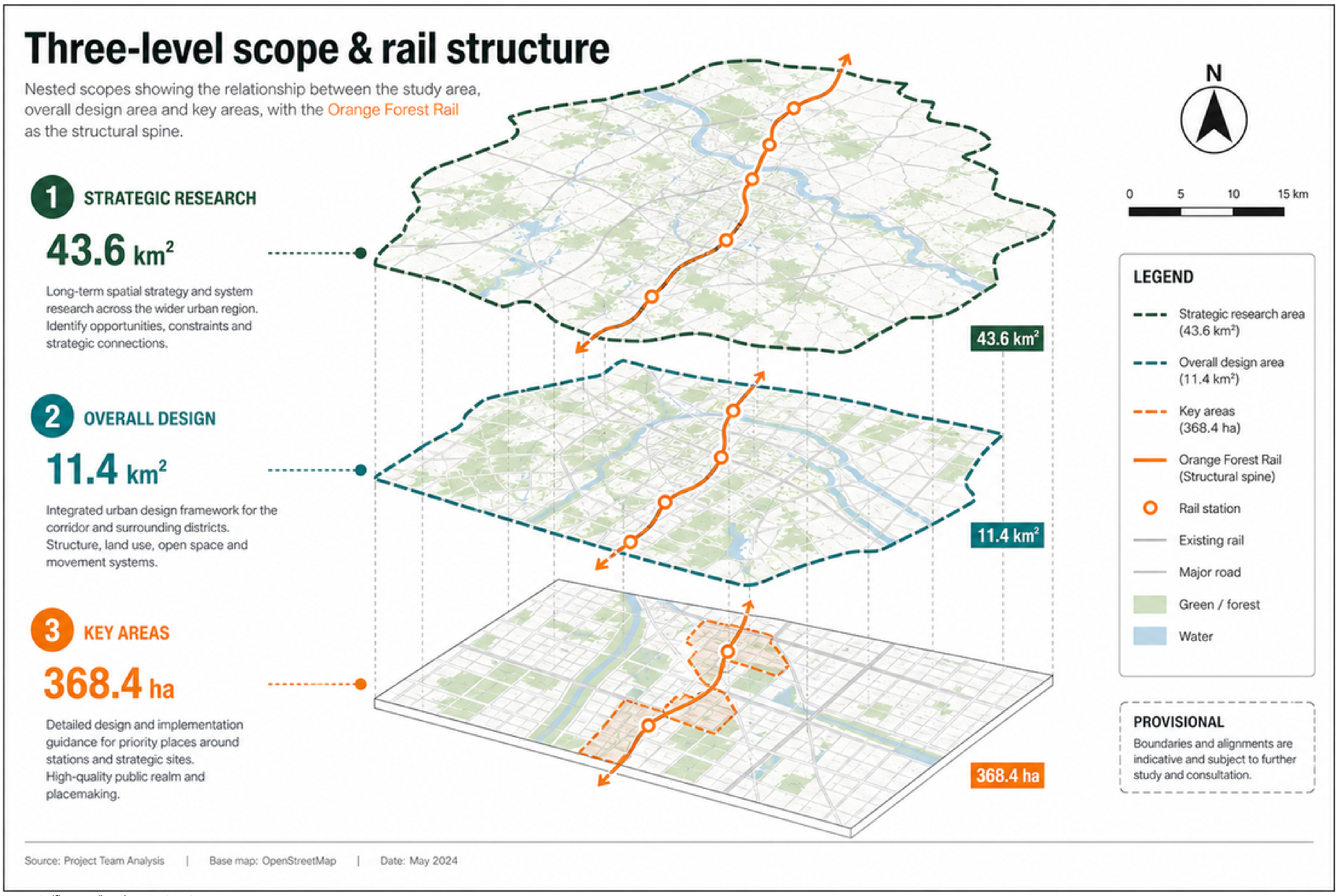
• data/processed/missing_data_checklist.csv

• Full source index: `sources.json`, `metrics.json`, `compliance_matrix.json`, `standard_matrix.json`, `design_depth_matrix.json`

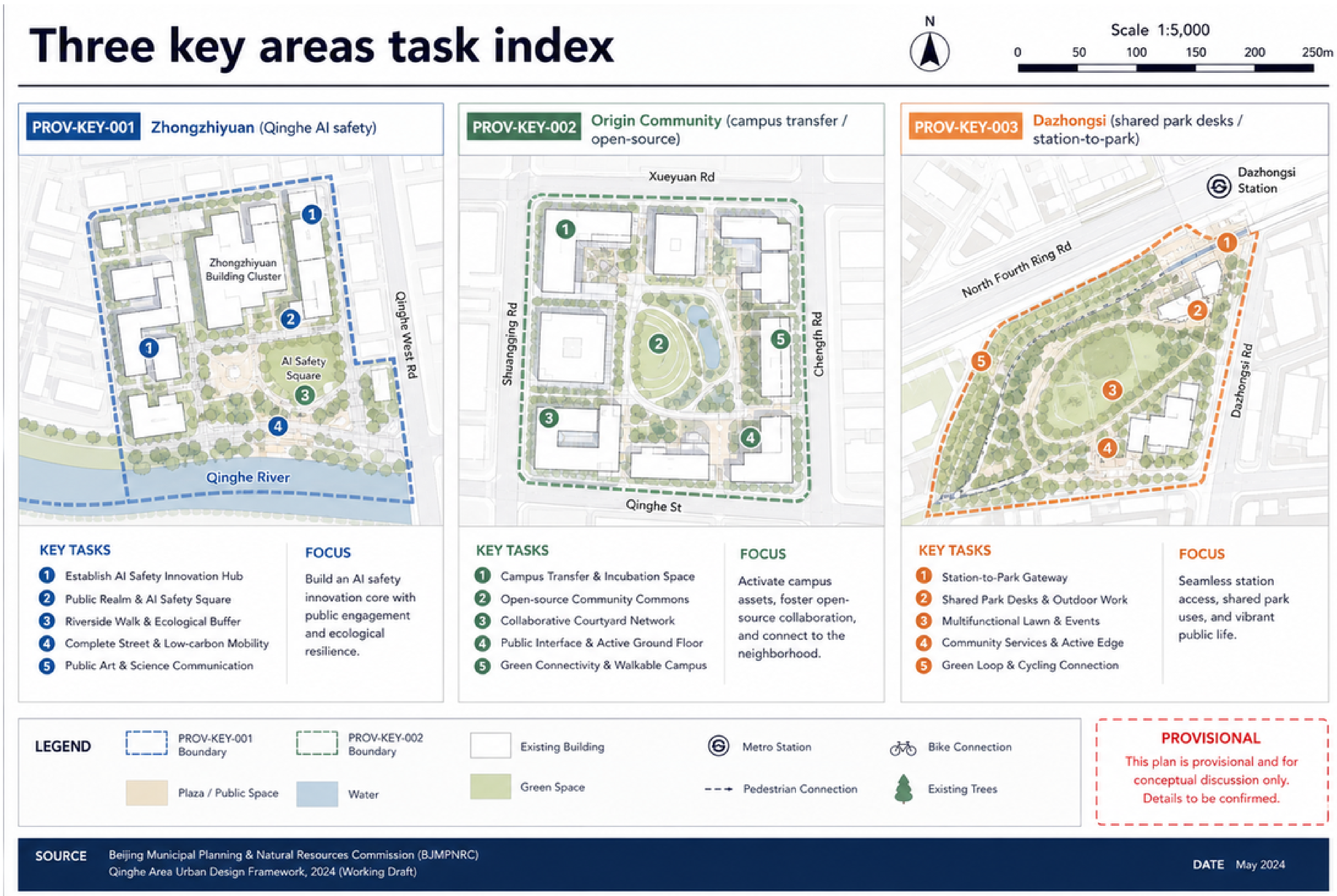
Evidence boards / 证据图



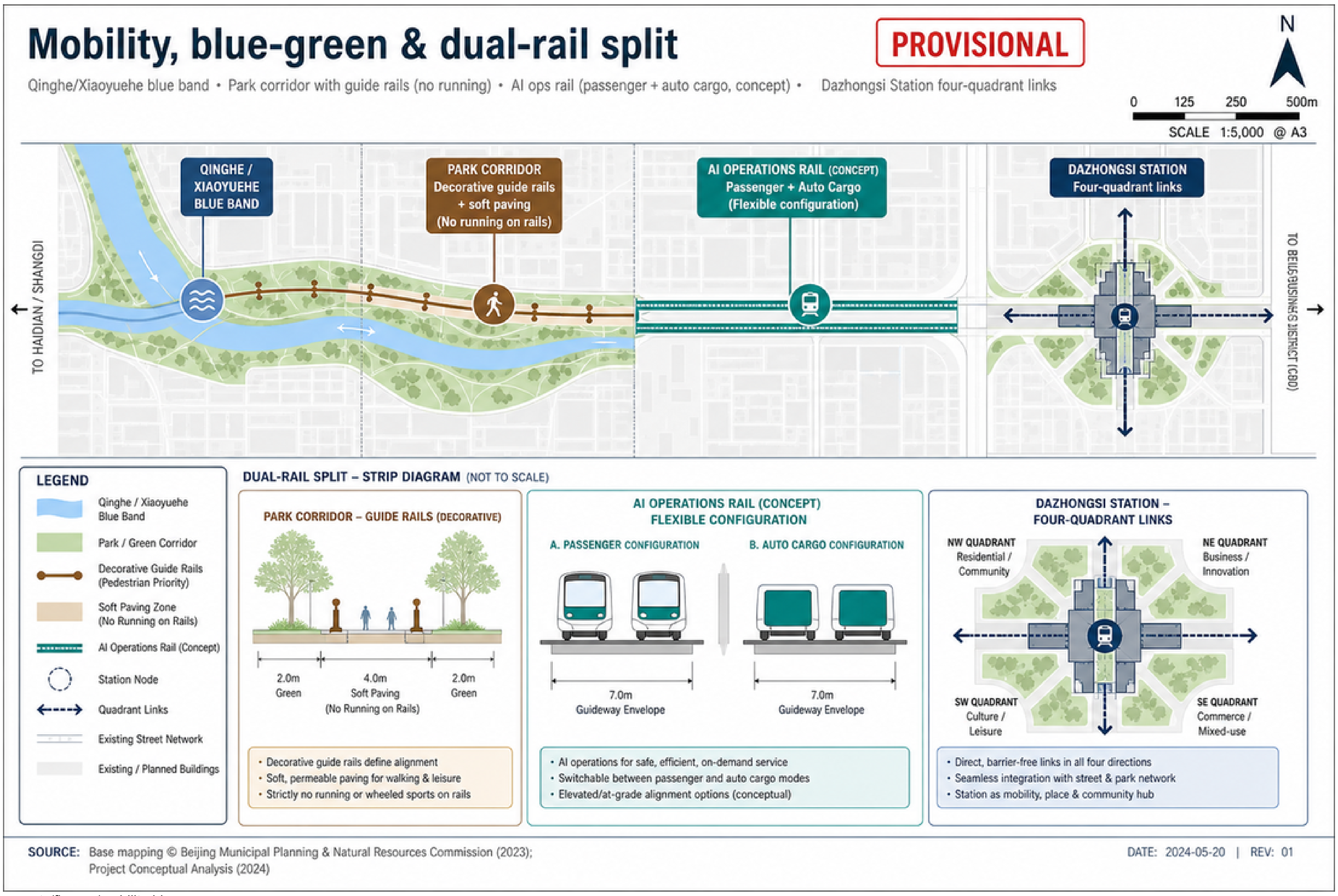
assets/figures/site-overview.en.png



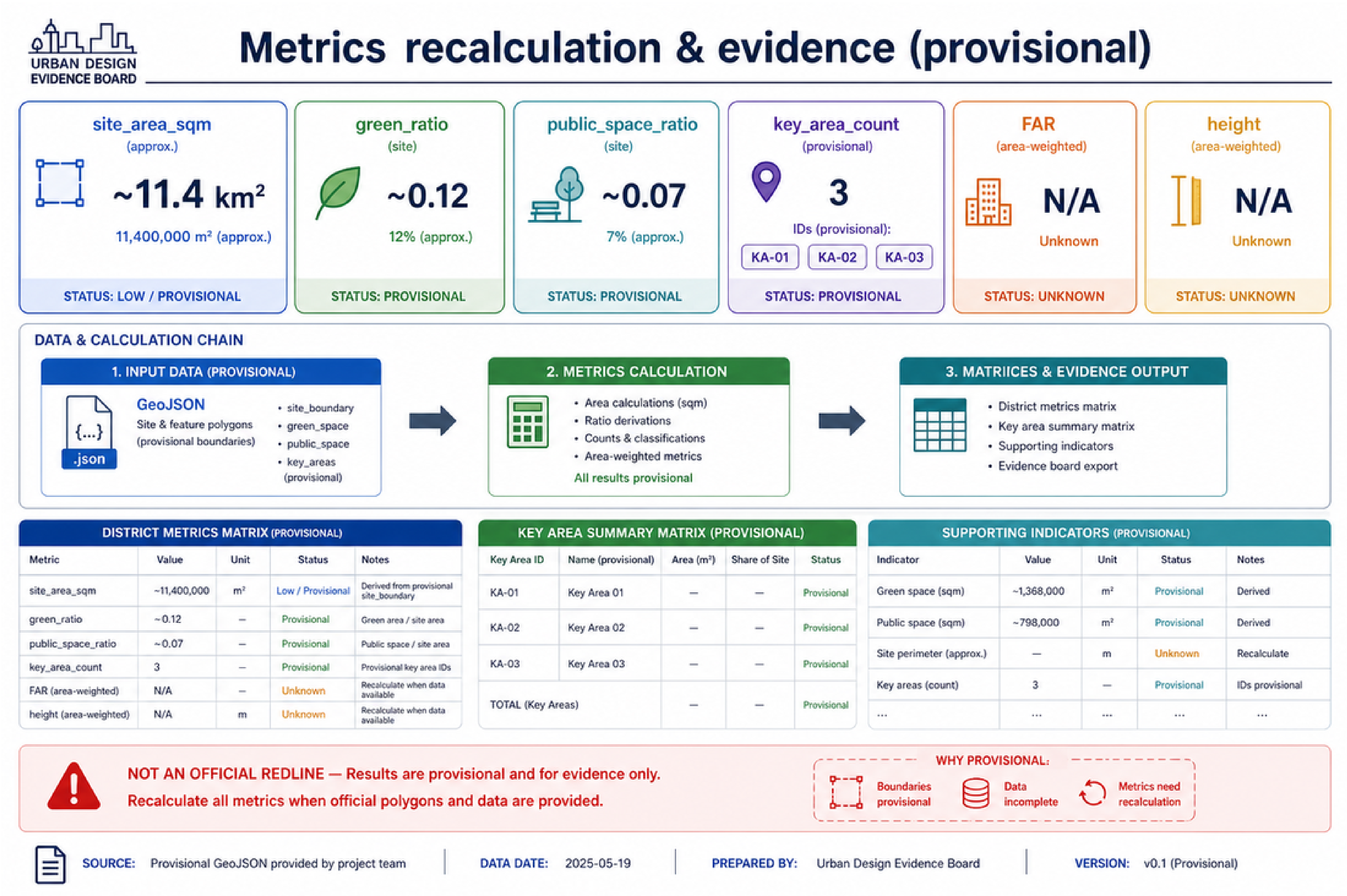

assets/figures/land-use-structure.en.png



assets/figures/key-areas.en.png




assets/figures/mobility-bluegreen.en.png




assets/figures/metrics-evidence.en.png